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AI Scientists Fail Without Strong Implementation Capability

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The emergence of Artificial Intelligence (AI) Scientist represents a paradigm shift in scientific discovery, with large language models (LLMs) taking the lead as the primary executor in the entire scientific workflow from idea generation to experiment implementation. Recent AI Scientist studies demonstrate sufficient capabilities for independent scientific discovery, with the generated research reports gaining acceptance at the ICLR 2025 workshop and ACL 2025, arguing that a human-level AI Scientist, capable of uncovering phenomena previously unknown to humans, may be imminent. Despite this substantial progress, AI Scientist has yet to produce a groundbreaking achievement in the domain of computer science on par with automated scientific tools. Based on extensive quantitative evidence from existing benchmarks in complex engineering tasks and a systematic evaluation assess 28 research papers generated by five advanced AI Scientist systems, we argue that **the fundamental bottleneck for AI Scientists lies in their capability to execute the requisite verification procedures.** Current AI Scientist systems lack the execution capabilities needed to execute rigorous experiments and produce high-quality scientific papers. To better illustrate the root cause of this **implementation gap**, we provide an in-depth discussion on the fundamental limitations of AI Scientist. This position paper aims to call for the participants in the community to bridge the implementation gap.

Keywords: AI Scientist, Implementation Gap, Hypothesis and Verification.

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Projects: https://ai-researcher.net
Code Repository: https://github.com/ResearAI/Awesome-AI-Scientist
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1. Introduction

The automation of scientific discovery has long been one of humanity's deepest desires (Langley, 1987, King et al., 2009, Radensky et al., 2024, AI, 2025). In recent years, with the advances in deep neural network technology, a range of **automated scientific tools** has emerged, leading to groundbreaking achievements in fields such as biomedicine (Yang et al., 2025c, Jumper et al., 2021), chemistry (Stokes et al., 2020), and materials science (Szymanski et al., 2023). For instance, DeepMind's AlphaFold can determine the 3D structures of proteins in just a few hours, a task that previously took years to solve (Jumper et al., 2021). In recent, researchers developed an autonomous laboratory, A-Lab, which successfully synthesizes 41 novel inorganic materials within 17 days (Szymanski et al., 2023). However, these scientific tools still rely heavily on human involvement. Researchers must first formulate ideas to be tested, while AI is responsible for the

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labor-intensive tasks of verification and iterative search. Therefore, these systems cannot be considered as truly automated scientific research.

The emergence of LLM-based **AI Scientist** has propelled the automation of scientific research to the next level, with AI taking the lead as the primary executor of scientific discovery, managing the entire workflow from idea generation to experiment execution (Lu et al., 2024, Weng et al., 2025). Recent studies have shown that research papers produced by AI Scientist have already reached the level of submissions to major machine learning conferences (Si et al., 2024, Yamada et al., 2025, Intology, 2025). As shown in Figure 1, we demonstrate the progress made by AI Scientist-v2 (Yamada et al., 2025), and the research output has received review scores exceeding the average acceptance threshold for human-authored papers. Similarly, researchers present an empirical validation through multiple peer-reviewed publications accepted at ICLR 2025 workshops and ACL 2025 main conference (Intology, 2025). Despite this substantial progress, AI Scientist has yet to produce a groundbreaking achievement in the domain of computer science on par with automated scientific tools (e.g., AlphaFold (Jumper et al., 2021)).

In this position paper, we first propose a conceptual framework (Section 2) that defines an AI Scientist as **an advanced end-to-end system capable of independently formulating scientific ideas and performing the implementation for verifying these ideas.** This definition forms the theoretical foundation of our position, aligns with current research progress (Lu et al., 2024, Weng et al., 2025, Yamada et al., 2025), and emphasizes that the core capability of an AI Scientist lies in generating innovative and feasible ideas at scale (Si et al., 2024, Wang et al., 2024a, Hu et al., 2024, Yang et al., 2025d). The idea-generation capability is a key feature that sets AI Scientists apart from automated scientific tools. While recent advances demonstrate that AI Scientists can generate highly innovative ideas (Si et al., 2025), their implementation capabilities remain constrained (Chan et al., 2024, Starace et al., 2025, Xiang et al., 2025, Siegel et al., 2024, Padigela et al., 2025), creating a significant gap between innovative idea generation and complete implementation.

> ***Our Position:*** *The fundamental bottleneck for AI Scientists lies in their implementation capability to effectively execute the verification of these ideas.*

We defend our argument by analyzing quantitative evidence from existing benchmarks used to evaluate LLMs' abilities in performing complex engineering tasks (Section 3.2). While LLMs can generate highly novel ideas (Si et al., 2024, Chai et al., 2024, Gottweis et al., 2025), their performance in experiment execution is exceptionally poor (Table 1). For instance, a leading LLM like Claude 3.5 Sonnet scored only **1.8%** on PaperBench (Starace et al., 2025). This **implementation gap** is further supported by a systematic evaluation (Section 3.3), which leverages a state-of-the-art review model, DeepReviewer-14B (Zhu et al., 2025), to assess 28 research papers generated by five advanced AI Scientist systems. The results demonstrate that current AI Scientist systems lack the execution capabilities needed to execute rigorous experiments and produce high-quality scientific papers. Finally, to clearly illustrate the root cause of the implementation gap, we provide an in-depth discussion on the fundamental limitations of AI Scientist (Section 4).

In summary, this paper validates and deeply analyzes the implementation gap in existing AI Scientist systems based on extensive quantitative evidence and a simulated peer-review process. Furthermore, as the development of AI Scientists will bring greater regulatory challenges, we comprehensively examine the ethical considerations (Section 5) faced by AI Scientists and suggest directions for future research (Section

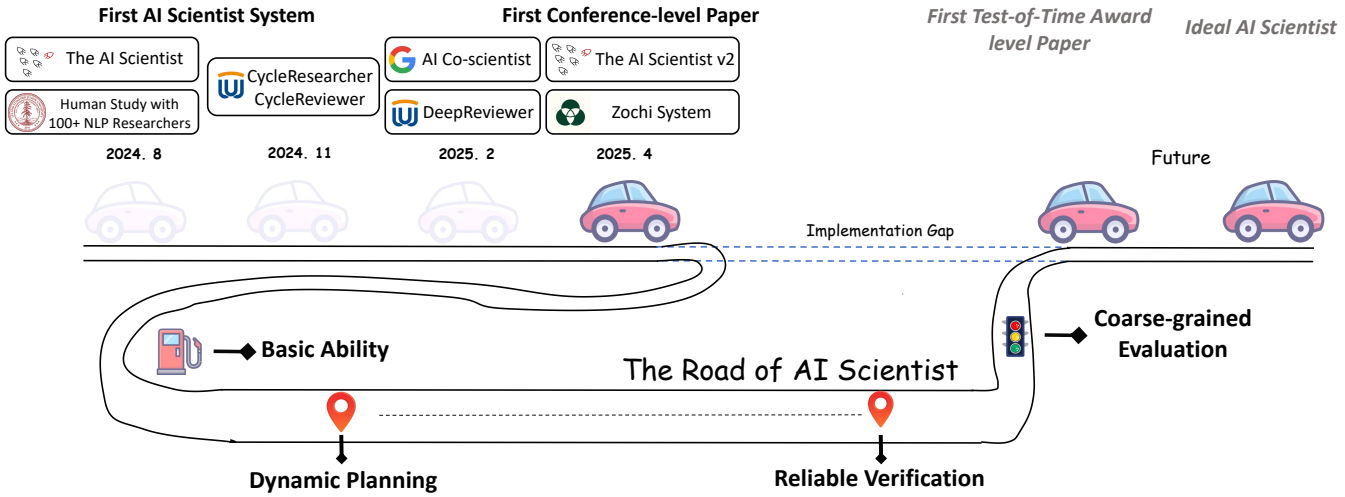


Figure 1: The roadmap of AI Scientist from 2024 to future, highlighting key milestones and fundamental challenges that must be overcome to bridge the implementation gap of AI Scientist.

6). We hope this position paper will contribute to a clearer understanding of the limitations of current AI Scientist, shedding light on the future development of AI Scientist.

2. Definition of the AI Scientist

The emergence of automated scientific tools has accelerated scientific discovery across numerous domains (King et al., 2009, Yang et al., 2025c, Jumper et al., 2021, Stokes et al., 2020, Szymanski et al., 2023). However, these tools fundamentally operate within a paradigm where human researchers remain in the dominant position of scientific discovery, and thus cannot be classified as fully automated AI Scientists. In this section, we first provide a detailed discussion of the unique characteristics of the AI Scientist (Section 2.1). Building on this discussion, we then propose a conceptual framework that formally defines the AI Scientist in a mathematical form (Section 2.2).

2.1. Unique Characteristics

Scientific tools, originating from AI for Science research, represent specialized AI systems designed to solve specific scientific problems by processing data and generating results within defined domains. These tools have demonstrated remarkable success across diverse scientific fields, including protein structure prediction (e.g. AlphaFold) (Jumper et al., 2021), antibiotic discovery through deep learning approaches (Stokes et al., 2020), and autonomous chemical research with large language models (Boiko et al., 2023). These scientific tools funda-

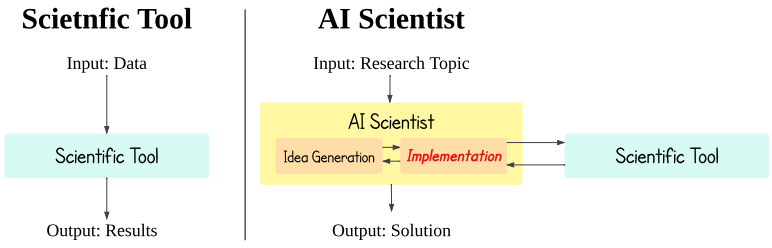


Figure 2: The difference between AI Scientist and scientific tool. Scientific tools generate predictions under human supervision, while AI Scientist autonomously leverage tools to address research questions.